

Technical Screening Task

Full-Stack Developer

Thank you for taking the time to participate in this challenge. Please read carefully the instructions below. This exercise is a simple order manager. You should develop a web app where users can create and manage orders. Items can be ordered and orders are automatically fulfilled as soon as the item inventory allows it.

Specification

The system should be able to provide the following features:

- Create, read/list, update and delete all entities.
- When an order is created, it should try to satisfy it with the current inventory.
- When an inventory movement is created, the system should try to attribute it to an order that isn't complete.
- When an order is complete, send a notification by email to the user that created it.
- Trace the list of inventory movements that were used to complete the order, and vice-versa.
- Show current completion of each order.

Entities and Attributes

Item	name
InventoryMovement	creationDate, Item, quantity
Order	creationDate, Item, quantity, User (who created the order)
User	name, email

Requirements

- The app should have a web frontend and a java (or other technology) backend.
- Use the programming language and tools that you feel more comfortable with.
 - We provide you with a boilerplate project that should ease the development, but feel free to use other tools you feel are more adequate.
- You should provide detailed instructions on how to run the project. Make sure we can run it to properly evaluate it.

You should

Provide rough estimates of how much time you'll need to develop this (core development, quality assurance) before starting the project.

Environment Setup

In order to ease the environment setup, we provide a development environment with Java, Play Framework and MySQL containers. Please read the README file in the supplied ZIP if you want to use this environment.